

Chiral Induced Spin Selectivity (CISS) Catalysts

Background

The *Chiral Induced Spin Selectivity (CISS)* effect has emerged as a fascinating quantum phenomenon that couple's electron spin with molecular chirality, leading to spin-polarized electron transport without external magnetic fields. This effect opens new possibilities for designing highly selective and energy-efficient electrocatalysts for key reactions such as CO₂ reduction (CO₂RR), oxygen evolution (OER), and hydrogen evolution (HER).

Understanding and harnessing CISS in catalytic materials can fundamentally transform electrocatalysis by improving reaction kinetics, selectivity, and stability through spin-controlled charge transfer.

Possible Focus Areas Include:

- Synthesis of chiral inorganic or hybrid catalysts (e.g., chiral metal oxides, chiral coordination complexes, or nanostructured catalysts).
- Surface modification of electrodes with chiral molecules or ligands.
- Electrochemical testing for CO₂RR, OER, or HER activity and selectivity.
- Characterization of catalysts with McAFM, XRD, EPR, SEM, TEM ...
- Correlation of spin polarization with catalytic performance.
- (Optional) First-principles or DFT-based simulation of spin-dependent charge transfer processes.

Your Profile

- Bachelor's degree in chemistry, Physics, Materials Science, Chemical Engineering, or a related field.
- Strong interest in catalysis, electrochemistry, or spintronics.
- Basic laboratory or simulation experience (e.g., electrochemical techniques, nanomaterial synthesis, or quantum chemistry).
- Motivation to work in an interdisciplinary environment bridging chemistry, physics, and materials science.
- Good communication skills in English (both written and spoken).

We Offer

- A highly interdisciplinary research environment within an international team.
- Access to state-of-the-art facilities for electrochemical testing and materials characterization.
- Opportunity to contribute to cutting-edge research in spin-dependent catalysis.
- Close supervision and the possibility to publish results in peer-reviewed journals.

Application

Interested candidates are invited to submit the following documents in a single PDF file:

- Motivation
- Curriculum Vitae (CV)
- Contact information of one academic reference

Application deadline: December 20,2025

Send your application at: leta-takele.menisa@chemie.uni-halle.de